OPERATING SYSTEMS

UE24CS242B

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File Management

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OPERATING SYSTEMS

AWK and SED

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Slides Credits for all PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau
5. Internet source

Text Processing Tools



Used for:

- Processing text files
- Data extraction & transformation

Common tools:

- **awk**
- **sed**

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AWK – Introduction & Working



- Scripting language for:
  - **Pattern scanning & processing**
- Works line-by-line
- Used for:
  - Data filtering
  - Report generation

Working

- Reads file line by line
- Splits line into **fields**
- Applies:
  - Pattern → Action

AWK - Syntax

Syntax:

- awk 'pattern {action}' file

Example:

- awk '{print $1,$4}' employee.txt

AWK – Important Operations



- Field access → $1, $2, …
- Conditions → if
- Built-in variables:
  - NR → number of lines
  - NF → number of fields

AWK – Selected Examples

awk '{print $1,$4}' file.txt

Print selected columns

awk '/manager/ {print}' file.txt

Pattern matching

awk 'END {print NR}' file.txt

Count lines

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SED – Introduction



- Stream editor for:
  - **Text transformation**
- Works in a **single pass**
- Used in pipelines

Syntax

sed 'command' file

Example:

sed 's/hello/world/' file.txt

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SED – Common Operations

Substitute → s/old/new/

Delete → d

Print → p

Insert / Append

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SED – Selected Examples



sed 's/hello/world/' file.txt

Replace text

sed '2d' file.txt

Delete line

sed -n '2p' file.txt

Print specific line

sed '4,6d' file.txt

Delete range

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AWK vs SED



Feature	AWK	SED
Type	Programming language	Stream editor
Focus	Data processing	Text editing
Works on	Fields	Lines
Complexity	Higher	Simpler



THANK YOU

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